

Education ERP Project - Completion Report

Client-facing status summary prepared from repository evidence and Railway pre-production findings. Prepared on 16 August 2026.

Overall status	Local Phases 0-7 are complete with evidence; Phase 8 staging automation is in progress.
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Production readiness	Not production-ready yet. Human PWA UAT, pilot UAT, production integrations and release security checks pending.
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Railway pre-production	Public guardian route is reachable, but P6.1 flow is blocked by missing published admission form seed data.
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Executive Summary

The Education ERP has moved beyond documentation and scaffolding. The repository now contains a custom Frappe application, local Docker-based platform setup, institution and academic masters, student identity and document foundations, admissions workflow foundations, fee and accounting foundations, applicant and student portal foundations, local automated tests, security controls, migration templates and portable staging artifacts.

The work completed so far is best described as a verified local pilot foundation. It proves the core architecture and business rules using synthetic data and fake provider adapters. It does not yet represent a production go-live because human user acceptance, production-sized migration rehearsal, real provider credentials, production infrastructure, monitoring, backups and external security review are still pending.

Completion Snapshot

Area	Client meaning	Status
Platform baseline	Pinned Frappe v16, ERPNext, Education, CRM, Payments and custom app install together.	Complete locally
Custom ERP app	university_erp contains domain DocTypes, APIs, integrations, portals and tests without modifying upstream apps.	Complete foundation
Institution and academics	University/campus/college/department, session, calendar, program, class, section, curriculum, intake, faculty and timetable foundations exist.	Complete locally
Student identity and documents	Guardian, identity issuance, duplicate review, consent, document scan, verification, replacement, expiry and privacy export foundations exist.	Complete locally
Admissions	CRM handoff, dynamic form version, draft save/resume, eligibility, merit, seat allocation, offer, confirmation and conversion foundations exist.	Complete locally
Fees and accounting	Fee policy, demand, invoice, online/offline payment, receipt, refund, settlement and GL reconciliation proofs pass.	Complete locally
Applicant/guardian PWA	Mobile bilingual portal, draft API, document scan and safe payment retry are implemented locally. Human testing remains.	Partially accepted
Student/guardian portal	Scoped access, dues, receipt PDF, documents, notices, fake OTP and payment status proof exist.	Complete locally
Security and migration	Audit/correlation, masking, private file limits, migration templates, checksum reconciliation and UAT scripts exist.	Complete local gate
Deployment automation	Railway, Hostinger Compose and AWS ECS artifacts exist; fresh image/deploy/monitoring/backups remain.	In progress

Work Completed By Phase

Phase	Completed result
P1	Local Frappe stack, site bootstrap and custom app installation verified.
P2	Pinned app compatibility, ERPNext accounting proof and fake provider adapter contracts completed.
P3	Institution, academic, student identity and document foundations completed with local migration/proof evidence.
P4	Admissions foundations completed: CRM handoff, application forms, eligibility, merit, seats, offers and conversion.
P5	Fee foundations completed: policy, demand, invoice, payment, receipt, refund, settlement and GL reconciliation.
P6	Applicant/guardian PWA and student/guardian portal implemented locally; human mobile/Hindi/usability checks deferred.
P7	Automated tests, security/privacy controls, migration validator, synthetic reconciliation and UAT scripts completed.
P8	Railway-first staging and portable production runtime artifacts started; deployment evidence remains incomplete.

Business Capabilities Now Proven Locally

- A pilot institution can be modeled with hierarchy, academic sessions, program offerings, classes, sections, curriculum, intake and faculty/timetable records.
- Applicant and student identity foundations support guardians, consent, immutable identity issuance, duplicate candidates, correction requests and privacy-oriented access logs.
- Admissions can move from CRM lead to application draft, eligibility evaluation, merit ranking, seat offer, confirmation and one-time student/enrollment conversion.
- Education fee demand generation reconciles to ERPNext Sales Invoice totals and supports concessions, scholarship, fines and waiver math in the local proof.
- Payment collection proofs show that duplicate webhooks and browser callbacks reuse the same accounting result instead of creating duplicate Payment Entries.
- Refund, settlement import and GL reconciliation proofs demonstrate balanced finance records with mismatch rejection.
- Portal APIs support draft resume tokens, private document scan state, application payment retry safety, student dues, receipt PDF download and status polling.
- Security controls include correlation IDs, masked restricted identifiers, approved privacy exports, private file URL TTL limits and webhook replay/signature checks.

Evidence And Verification

The evidence base is stored inside the repository under docs/evidence. The important point for the client is that these are not just screenshots or promises: the local gates include migrations, repeatable proof functions, automated tests, documentation checks and repository policy checks.

Evidence group	Verification highlights
Local platform	Docker Compose services, MariaDB health, site migration, app list, HTTP smoke tests and Desk route smoke tests.
Accounting and payments	Sales Invoice, Payment Entry, duplicate event reuse, credit note, refund Payment Entry and GL balance checks.
Provider adapters	Fake Razorpay, MSG91, SMTP, R2 and ClamAV success/failure/timeout/idempotency contracts.
Domain proofs	Repeatable proof functions for academic masters, identity/documents, admissions, fees and reconciliation.
Automated tests	10 custom-app tests passed locally by P8.1 evidence, including database/cache readiness and security controls.
Migration/UAT	CSV templates, no-write validator, immutable checksums, count/reference/finance reconciliation and nine UAT scenarios.

Railway Pre-Production Finding

The deployed Railway URL was tested during P6.1 preparation:

<https://web-production-7580e.up.railway.app/guardian-admission> returned HTTP 200. This means the public guardian route is reachable.

The first real P6.1 flow test failed at draft save. The public context endpoint returned forms:

[], and a correctly encoded synthetic save request returned: No published admission form is available. This is a pre-production data/configuration blocker, not proof that the whole portal is broken.

Required fix before continuing P6.1

- Log in to Frappe Desk on the Railway site as Administrator or an authorized academic/system user.
- Create or publish one Admission Application Form Version for the pilot program and academic year.
- Use a valid JSON form schema containing a fields list.
- Retest the English guardian registration flow after the context endpoint returns at least one published form.

Not Yet Complete

- The product is not approved for production, pilot go-live, real provider traffic or real-data import.
- P6.1 human mobile checks are still open: English full flow, Hindi review, offline recovery, document upload from phone, payment callback persistence and low-digital-literacy guardian test.
- P7.3 pilot UAT must still be run with named pilot users in actual roles, from UAT-001 through UAT-009.
- A production-sized masked migration rehearsal must still run on an isolated site with count, reference, fee, invoice, payment, outstanding and GL reconciliation sign-off.
- Production Razorpay, MSG91/DLT, Hostinger SMTP, Cloudflare R2, DNS, WAF, TLS, monitoring, backup/PITR and restore tests are not yet approved or completed.

- Production MFA, external security review, vulnerability/container scans, image signing, registry SBOM verification and final release manifest remain open.

Recommended Next Steps

Priority	Action	Owner / evidence needed
1	Fix Railway seed data by publishing one Admission Application Form Version.	Product/Academic admin; screenshot and API context response.
2	Resume P6.1 guardian PWA testing end to end at 320px, 375px and 390px plus a real phone.	QA/Product; dated evidence file with screenshots and defects.
3	Close Hindi text defects and upload/payment edge cases found during P6.1.	Engineering/QA; retest evidence.
4	Run pilot UAT-001 through UAT-009 with named users in their actual roles.	Institution/Product/Finance/Security; signed acceptance.
5	Run masked production-sized migration rehearsal on isolated staging.	Migration/Finance; checksum, count, reference and GL reconciliation.
6	Complete P8.1 staging hardening: image, SBOM, Railway services, backups, monitoring and restore test.	DevOps; registry/deploy/restore evidence.

Client Decision Summary

The completed work supports a clear client message: the project has a substantial verified foundation and many core ERP flows are implemented locally with synthetic proof data. The correct next commercial milestone is not production launch; it is pre-production stabilization and user acceptance on Railway/staging.

Recommended client-facing position: approve continuation into pre-production UAT after the Railway seed-data blocker is fixed, while keeping real provider traffic and real data blocked until all release rules and sign-offs are complete.